Lecture 26 (CA+CP+CV)

Tuesday, 18 April 2023 1:31 PM

HW #4 → Thursday

Final report → Friday

Exam 2 → 4/28 } HW material #4

HW 0 pt 2

Fundamental Techniques:

+ Chronoamperometry (CA)
+ Chronopotentiometry (CP)
+ Polarization curves (LSV, CV)
+ Electrochemical Impedence Spectroscopy (EIS)

→ CA's: Chrono → time, Amp → current } changing current (while holding potential constant)

Time dependent conc. profile can be solved w/ MT concepts

ex) Assume:

- Double layer current negligible (Since it takes time/charge for this double layer to form)
- No forced convection } Diffusion controlled
- $t_j \to 0$

Mass balance:

$$\frac{\partial c_j}{\partial t} = -\nabla \cdot N_j + R_j$$

($R_j \to 0$; N_j = migration + diff. + convection) } This may not be always valid (steady state voltammetry) → Application of current usually will usually result in gradients & "flow" → harder to reach steady state in these larger electrodes

PDE $\boxed{\frac{\partial c_j}{\partial t} = D \frac{\partial^2 c}{\partial x^2}}$

BC's: @ $t=0$; $c = c_\infty$

@ $x \to \infty$; $c = c_\infty$

@ $[t>0, x=0]$; $c = 0$ ← limiting current

Combination of variables or Laplace Transform to solve PDE:

$$\frac{c(x,t)}{c_\infty} = \frac{2}{\sqrt{\pi}} \int_0^{\frac{x}{\sqrt{4Dt}}} \exp[-y^2]\, dy$$

$$= \operatorname{erf}\left[\frac{x}{2(Dt)^{1/2}}\right]$$

from conc. profile we get flux to then get current (i)

Flux @ $x=0$:

$$\left.\frac{\partial c}{\partial x}\right|_{x=0} = \frac{2c_\infty}{\sqrt{\pi}} \exp\left[\frac{x^2}{4Dt}\right] \left.\frac{1}{\sqrt{4Dt}}\right|_{x=0}$$

$$= \frac{c_\infty}{\sqrt{\pi Dt}}$$

$$-N_{ox} = D\left.\frac{\partial c}{\partial x}\right|_{x=0} = \frac{D c_\infty}{\sqrt{\pi D t}}$$

$$\boxed{i(t) = \frac{nF}{s}\frac{D c_\infty}{\sqrt{\pi D t}}}$$ ← Cottrell Equation

Note: As: $t \uparrow$, $i \downarrow$ (depletion of reacting species near the surface) → : $t \to 0, i \to \infty$

↳ infinite driving force (potentiostat usually has current limits that will artificially cap current)

Since we are only considering diffusion (so the species close to the electrode will react & depleting this layer more & more)

* According to $c(x,t)$
↳ diffusion BL (defined as x @ $c = c_\infty$) grows w/ t.

CP's: Vary potential (constant current) w/ time } Same system as before

+ Material balance: $\frac{\partial c}{\partial t} = D\frac{\partial^2 c}{\partial x^2}$

@ $t=0$, $c = c_\infty$

@ $x \to \infty$, $c = c_\infty$

@ $x=0$, $t>0$, $i = \frac{nF}{s} D\frac{\partial c}{\partial x}$ [Flux BC]

↑ Constant!

* Will this result in $c_{lim}=0$ @ all $t>0$?
↳ no, initially $c_{lim}>0$, @ some time c_{lim} will get depleted @ electrode

* What happens after $c_{lim} \to 0$ when applying constant current? (Conc. of our limiting species goes to zero)
↳ Since we are applying a constant current, and all the species have been depleted
↓
Secondary reaction will turn on which may cause abrupt change in potential } Something else has to make up the current (pass charges)

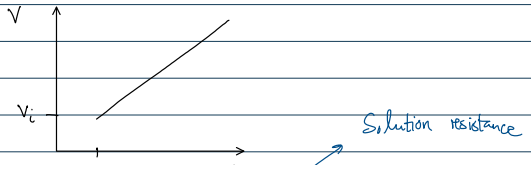
Sand Eq (for semi-infinite geometry)

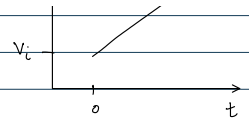
The time @ which $c_{lim} \to 0$ is given by solution to PDE w/ const. flux BC:

$$\boxed{\tau = \frac{\pi D}{4}\left(\frac{c_\infty n F}{s_{lim} q}\right)^2}$$

Polarization curves (LSV / CV)

→ A polarization (I vs. V) curve can be obtained by linear sweep voltametry (LSV)
(Apply linearly changing potential function + measure current)





Solution resistance

This measurement is affected by ohmic drop in solution
↳ potential gradient in solution resulting from current flow b/w WE & CE

$$\eta = (V - V_{ref}) - (U - U_{ref}) - \Delta\phi_{\Omega}$$

$(V - V_{ref})$: RE potential; $(U - U_{ref})$: ↙; $\Delta\phi_{\Omega}$: Ohmic drop

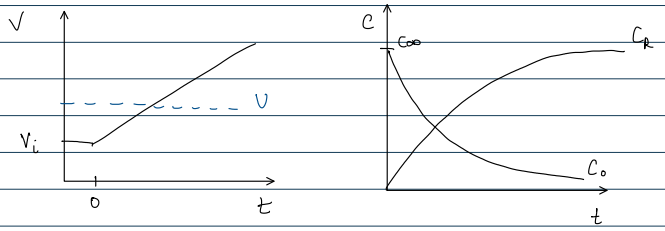
↳ We can estimate the value of $\Delta\phi_{\Omega}$ @ RE far away from disk electrode for primary current distribution

$$\Delta\phi_{\Omega} = \frac{\pi r_0}{4\kappa} i_{avg}$$

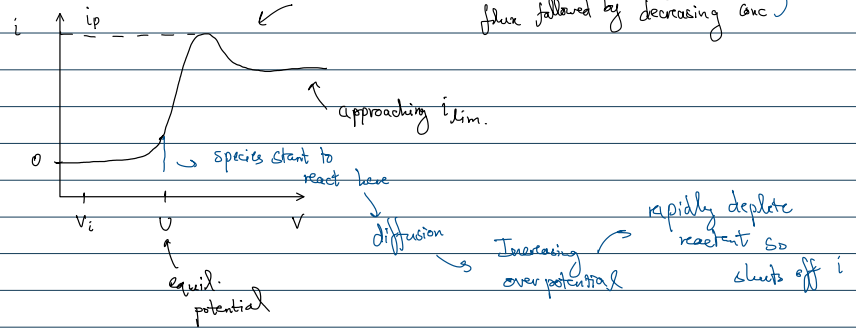
r_0: WE radius; ; Ohmic drop depends on electrode size & electrolyte conductivity [proportional to current]

↳ LSVs for reversible rxns, w/ only one half of redox couple in solution

ex: $O + ne^- \rightleftharpoons R$ w/ equil. potential U if solution contains O, but no R, and we apply an LSV



How does i vs. V look like?



we get a peak (non-SS) that results from increasing flux followed by decreasing conc } Non-SS process

approaching $i_{lim.}$

species start to react here

diffusion

Increasing overpotential

rapidly deplete reactant so shuts off i

For $V(t) = V_i + vt$ (No species O initially in the system)

v: Scan rate $(\frac{V}{s})$

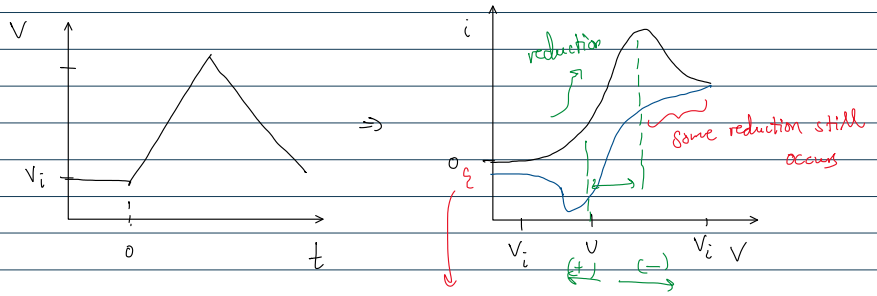
$$i_p = -0.4463\,(nF)^{3/2}\left(\frac{D_O}{RT}\right)^{1/2} C_{OO}^{*}\, v^{1/2}$$ → † For semi-infinite (like disk electrode)

† Diffusion only

Height changes but not the peak (as you sweep faster)

Often we want to reverse the potential sweep to create a cyclic voltammogram (CV)

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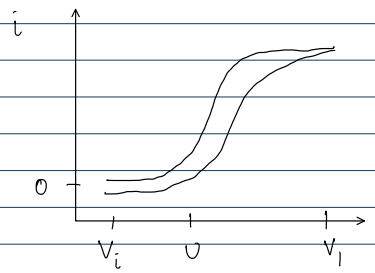


We don't reach V_i because we do have species that can be reduced

Comes from the math } ~60 mV (for reversible process) $\left[\frac{2.303RT}{F}\right]$

These techniques have complex math analysis due to time response but we can simplify some of these techiques by using "ultramicro electrodes" (UME)

↳ electrodes ~10μm → (establish SS in ecem synthesis @ much shorter time scales than standard electrodes)



No peaks!

(SS voltammetry)